Mid-term Exam Review

Science & Technology Innovation Management

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Exam: 23/10/2026, 14:55–15:55 (written, 1 hour)

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1. Definition of innovation

EN: Innovation is the successful exploitation of new ideas – turning an invention (a new idea or technology) into practical, commercially or socially useful value. It is distinct from invention (creating something new) and from diffusion (the later process by which the market adopts it).

VN: Đổi mới sáng tạo (innovation) là việc khai thác thành công một ý tưởng mới – biến một phát minh (ý tưởng hoặc công nghệ mới) thành giá trị thực tế, có ích về mặt thương mại hoặc xã hội. Nó khác với sáng chế (invention – tạo ra cái mới) và khác với khuếch tán (diffusion – quá trình thị trường chấp nhận nó sau đó).

2. The "4Ps" model (Tidd & Bessant)

Product: what the organization offers — *Những gì tổ chức cung cấp*
Process: how it is made and delivered — *Cách tạo ra và phân phối sản phẩm*
Position: who/where it is sold to — *Bán cho ai, ở thị trường nào*
Paradigm: the underlying business model — *Chính mô hình kinh doanh nền tảng*

Apple. Product: iPhone, Apple Silicon chips designed in-house. Process: vertically integrated hardware, software, and chip design. Position: premium positioning, closed ecosystem. Paradigm: shift from one-time hardware sales to recurring Services revenue.

Viettel. Product: its own Open RAN 5G equipment, co-developed with Qualcomm, trialed with a UAE operator in 2025. Process: in-house R&D via Viettel High Tech. Position: exporting telecom equipment, competing with Huawei, Ericsson, Nokia. Paradigm: shift from telecom operator to a technology & defense conglomerate.

FPT. Product: FPT AI Factory (Nov 2024, Nvidia GPU-based AI cloud) and the first "Make in Vietnam" chip shipped to Japan. Process: building its own AI infrastructure instead of only staffing outsourced coding. Position: $200M investment

targeting Japan as an AI partner market. Paradigm: shift from software-outsourcing vendor to AI/digital-transformation partner, targeting $5B global revenue by 2030.

3. Risks in the "Implementing" phase

Technological risk *Rủi ro công nghệ*

The concept may not scale, may be too costly to manufacture, or may simply not work reliably outside the lab.

Market risk *Rủi ro thị trường*

Customers may not want it, demand may be smaller than expected, or a competitor/substitute may arrive first.

Managing both: stage-gate reviews (kill or fund at each checkpoint); build a minimum viable product/prototype to test technical feasibility cheaply; run a small pilot with real customers to validate demand before full-scale launch; validate technical and market risk in parallel, not sequentially.

VN: Dùng mô hình stage-gate (dừng hoặc cấp vốn tiếp ở từng mốc), làm MVP/prototype để kiểm tra tính khả thi kỹ thuật với chi phí thấp, chạy thử với khách hàng thật để kiểm chứng nhu cầu trước khi tung ra quy mô lớn, và kiểm chứng kỹ thuật lẫn thị trường song song thay vì lần lượt.

4. The role of innovation management (5 official roles)

1. **Creating & sustaining competitive advantage** – *tạo và duy trì lợi thế cạnh tranh liên tục*
2. **Integrating technology, market & organization** – *gắn kết công nghệ, thị trường, tổ chức*
3. **Transforming ideas into real value** – *biến ý tưởng thành giá trị thật, không chỉ ra mắt*
4. **Building & optimizing dynamic capabilities** – *xây và tối ưu năng lực thích ứng*
5. **Managing risk & optimizing resources** – *mở rộng nhanh mà không phình rủi ro*

Example – Grab, all 5 roles in one company:

1. Competitive advantage: EXP, hundreds of A/B tests a month tuning matching/pricing – why no rival, including Uber, out-executed Grab in Southeast Asia.
2. Integration: the super-app flywheel – EXP technology, 47M users, and licenses across 8 countries fused into one loop.

3. Ideas → value: GrabMaps, built in 2022 for its own drivers, later sold B2B; 14 years from MyTeksi to Grab's first net profit, $200M in FY2025.
4. Dynamic capabilities: incentive spend cut from 13.3% of GMV (2022) to about 10% (2024) while users kept growing.
5. Risk & resources: GrabDefense protected 4.5 billion transactions from fraud in 2024; years spent securing financial licenses before launching lending.

Good luck this afternoon, 14:55–15:55! / Chúc may mắn chiều nay!